Swarup K Subudhi

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Mechanical Engineer (Ph.D., Dec 2026) with four years taking defense and commercial transmissions from design review to mass production at Tata Motors, and four years designing, building and qualifying electro-mechanical hardware for NASA-sponsored programs. Strong in structural/thermal FEA, materials, DFM/DFA and harsh-environment testing.

ENGINEERING EXPERIENCE

Graduate Research Assistant, University of Maryland, College Park, MD — 2022 – present

3D High-Energy X-Ray Polarimeter Assembly via Additive Manufacturing (Sponsor: NASA-GSFC) — Jan 2024 – present

- Designed and prototyped a mechanism converting a 3-DOF extrusion printer to 5-DOF, integrating the motion control software, so electronics can be printed onto curved and irregular substrates.
- Developed direct-write extrusion processes achieving ±10 µm tolerance on 50–100 µm-pitch silver interconnects for a compact X-ray polarimeter assembly; verified geometry by optical/laser metrology and surface profiling.
- Wrote vibration qualification test matrices to NASA GEVS launch-load profiles and set pass/fail criteria; calibrated accelerometers and strain gauges (noise floor) before shaker runs; ran thermal and humidity exposure tests.

Thermal Cycling Reliability of Printed Electronics — Jun 2023 – Dec 2023

- Built parametric multilayer FEA models (ANSYS Mechanical, Abaqus) of dielectric/sintered-silver stacks under −40 to +140 °C cycling (JEDEC JESD22-A104); an Anand viscoplastic model located crack initiation sites that matched SEM cross-sections.
- Measured CTE, modulus and creep inputs with TMA, DMA and DSC; applied physics-of-failure and acceleration-factor analysis.

Dissertation: Magnetic Films & Drops in Viscous Media — Jan 2022 – present

- Designed and built electro-mechanical test setups to actuate magnetic fluids and films; calibrated VSM and Hall-effect magnetometers for magnetic field mapping.
- Built Python/MATLAB pipelines that extract geometry and motion from large high-speed image sets.

Senior Manager, Manufacturing (Transmissions), Tata Motors CVBU, Jamshedpur, India — 2019 – 2020

- Led a cross-functional Six Sigma team (NPI, QA, supply chain, R&D) that cut gearbox bearing failures ~50% within the 24-month warranty window across the G600/750 and G1150 families, saving ₹4.53 Cr (≈$0.6M) a year on one truck line; used DoE, Gauge R&R and SPC.
- Raised gearbox assembly-line efficiency ~12% by rebalancing takt time, adding PFMEA-driven quality checks, redesigning tooling and fixture heights after ergonomic (RULA) studies, and deploying an IoT ANDON loss-capture system.
- Value engineering: new material and surface treatment for 9-speed gearbox synchronizer cones to raise durability.

Assistant Manager, New Product Introduction (Transmissions), Tata Motors CVBU, Jamshedpur, India — 2016 – 2019

- Led NPI of the auxiliary gearbox for the Tata Kestrel 8×8 amphibious combat vehicle from design review through proto build, supplier development, DFA/DFM, assembly and integration testing; owned the performance, NVH and durability test matrix through Army acceptance.
- Launched 10 gearbox variants, including 6×6 and 8×8 defense transmissions, into rate production with 100% on-time delivery; ran first-article inspections, dimensional audits, Cpk studies and engineering change control in PLM.

EDUCATION

Ph.D., Mechanical Engineering, University of Maryland, College Park (GPA 3.83) — Dec 2026 (expected)

M.S., Mechanical Engineering, University of Maryland, College Park — 2021

B.Tech., Mechanical Engineering, National Institute of Technology, Rourkela, India — 2016

TECHNICAL SKILLS

CAD & design: SolidWorks (assemblies, drawings), Creo; GD&T, DFM/DFA, PLM / ECN
Structural & thermal analysis: ANSYS Mechanical, Icepak, Abaqus (nonlinear, viscoplastic, creep), COMSOL, Fluent
Electro-mechanical: Arduino, stepper/servo control; strain gauges, load cells, encoders, accelerometers, Hall sensors
Materials & testing: TMA, DMA, DSC, SEM/EDS, metallography; vibration (GEVS), thermal cycling (JEDEC), Weibull
Manufacturing & quality: NPI, PFMEA, Six Sigma, DoE, SPC/Cpk, MSA; CNC, FDM/SLA, direct-write; Python, MATLAB, C++
Publications: 3 first-author journal papers (*ACS Appl. Mater. Interfaces*, *Flex. Print. Electron.*, *Front. Sens.*) and 4 co-authored